DAV SCHOOL

SREE NANDEESWARAR CAMPUS

ADAMBAKKAM, CHENNAI-88

CLASS : IX **PHYSICS**

CHAPTER 10 – GRAVITATION

PART B – FREE FALL

Answer the following questions

1. What do you mean by free fall?

The falling of object towards the Earth only due to the gravitational force of earth is called free fall.

2. What do you mean by acceleration due to gravity?

The uniform acceleration produced in a free falling body due to the gravitational force of the earth is called as acceleration due to gravity. It is represented as (g). $g = 9.8 m/s^2$

3. Distinguish between mass and weight

MASS

- It is the quantity of matter contained in an object
- It is a vector quantity
- S.I unit is kg
- Mass of an object is always constant.
- It can never be zero
- It is measured with a beam or digital balance.

WEIGHT

- Weight is the force with which an object is attracted to the centre of the earth.
- It is a vector quantity.
- S.I unit is N
- Weight varies from place to place due to variation in 'g'.
- It can be zero. (eg) At the centre of the earth.
- It is measured using spring balance.

4. Why will a sheet of paper fall slower than one that is crumbled into a ball?

This is because air offers resistance to the sheet of paper due to friction which is more than the resistance offered by the air to the paper ball as the paper sheet has larger area.

5. Amid buys few grams of gold at the poles as per the instruction of one of his friends. He hands over the same when he meets him at the equator. Will the friend agree with the weight of gold bought? If not why?

The value of "g" is maximum at the poles and minimum at the

Equator.

$g_p > g_e$

Weight of an object at poles $W_p = m\, g_p$

Weight of an object at equator $W_e = m\, g_e$

ie, weight of the gold at the equator will be less than the weight of the gold at the poles. Obviously the friend at the equator will not agree.

6.

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Why is weight of an object on the moon $\frac{1}{6}$ th its weight on the earth?

Let mass of an object = m

Weight on Earth

Let

Mass of earth = M_E

$= 5.98 \times 10^{24}$ kg

Radius of earth = R_E

$= 6.37 \times 10^{6}$ m

,

Weight on earth = W_E

$$= \frac{GM_E\, m}{{R_E}^2}$$

Weight on Moon

Let

Mass of moon = M_M

$= 7.36 \times 10^{22}$ kg

Radius of moon = R_M

$= 1.74 \times 10^{6}$ m

Weight on moon = W_M

$$= \frac{GM_M\, m}{{R_M}^2}$$

Now

$$\frac{Weight\ on\ moon}{Weight\ on\ earth} = \frac{W_M}{W_E}$$

$$= \frac{GM_M\ M}{{R_M}^2} \div \frac{GM_E\ M}{{R_E}^2}$$

$$= \frac{GM_M\ M}{{R_M}^2} \times \frac{{R_E}^2}{GM_M\ M}$$

$$= \frac{M_M \times {R_E}^2}{M_E \times {R_M}^2}$$

Putting values

$$= \frac{7.36 \times 10^{22} \times \left(6.37 \times 10^6\right)^2}{5.98 \times 10^{24} \times (1.74 \times 10^6)^2}$$

$$= \frac{7.36 \times 10^{22} \times (6.37)^2 \times 10^{12}}{5.98 \times 10^{24} \times (1.74)^2 \times 10^{12}}$$

$$= \frac{7.36 \times (6.37)^2}{5.98 \times 10^2 \times (1.74)^2}$$

$$= \frac{7.36 \times 40.57}{5.98 \times 100 \times 3.02}$$

$$= \frac{298.59}{1805.9}$$

$$\approx \frac{1}{6}$$

Thus,

$W_M = \frac{1}{6} W_E$

∴ Weight of an object on the moon is $\frac{1}{6}$ of it's weight on the earth.

7. Write three equations of motion for freely falling bodies.

General equation of motion	Equations of motion for freely falling bodies
(i) $v = u + at$ changes to	$v = u + gt$
(ii) $s = ut + \frac{1}{2}at^2$ changes to	$h = ut + \frac{1}{2}gt^2$
(iii) $v^2 = u^2 + 2as$ changes to	$v^2 = u^2 + 2gh$

8.

NCERT Question 12

Gravitational force on the surface of the moon is only $\frac{1}{6}$ as strong as gravitational force on the earth. What is the weight in newton's of a 10 kg object on the moon and on the earth?

Given

Gravitational force on surface of moon

$$= \frac{1}{6} \times \text{Gravitational force on surface of earth}$$

Therefore,

$$\text{Weight on moon} = \frac{1}{6} \times \text{Weight on earth}$$

Finding Weight on Earth

Mass of the object = m = 10 kg

Acceleration due to gravity g = 9.8 m/s^2

Weight of object on earth = mg

$= 10 \times 9.8$

$= 98$ N

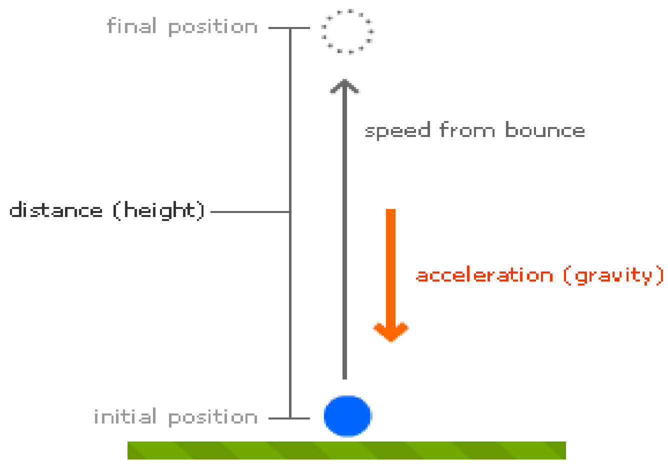
And,

$$\text{Weight on moon} = \frac{1}{6} \times \text{weight on earth}$$

$= \frac{1}{6} \times 98$

$= 16.33$ N

9. A ball is thrown vertically upwards with a velocity of 49 m/s. Calculate (1) The maximum height to which it rises. (2) The total time it takes to return to the surface of the earth.



Given parameters:

- Initial velocity of the ball (u) = 49m/s.
- The velocity of the ball at maximum height (v) = 0.
- g = $9.8 m/s^2$
- Let us considered the time taken is t to reach the maximum height h.

To find:

- (1) The maximum height to which it rises.
- (2) The total time it takes to return to the surface of the earth.

Solution:

Consider the formula,

- $2gh = v^2 – u^2$
- $2 \times (- 9.8) \times h = 0 – (49)^2$
- $– 19.6\ h = – 2401$
- **h = 122.5 m**
- Now consider the formula,

- v = u + g × t
- 0 = 49 + (- 9.8) × t
- – 49 = – 9.8t
- **t = 5 sec**
- **(1) The maximum height to ball rises = 122.5 m**
- **(2) The total time ball takes to return to the surface of the earth = 5 + 5 = 10 sec.**

10. A stone is released from the top of a tower of height 19.6m. Calculate its final velocity just before touching the ground.

Given:

- Initial Velocity u=0
- Height, h=19.6 m

TO FIND : Final velocity v=?

SOLUTION:

- By third equation of motion
- $v^2 = u^2$ + 2gh
- v^2 = 0 + 2×9.8×19.6
- v^2 = 384.16
- ⇒ v = 19.6 m/s

11. A stone is thrown vertically upward with an initial velocity of 40 m/s. Taking g = 10 m/s^2, find the maximum height reached by the stone. What is the net displacement and the total distance covered by the stone?

Given:

- Initial Velocity u = 40 m/s
- Final velocity v = 0

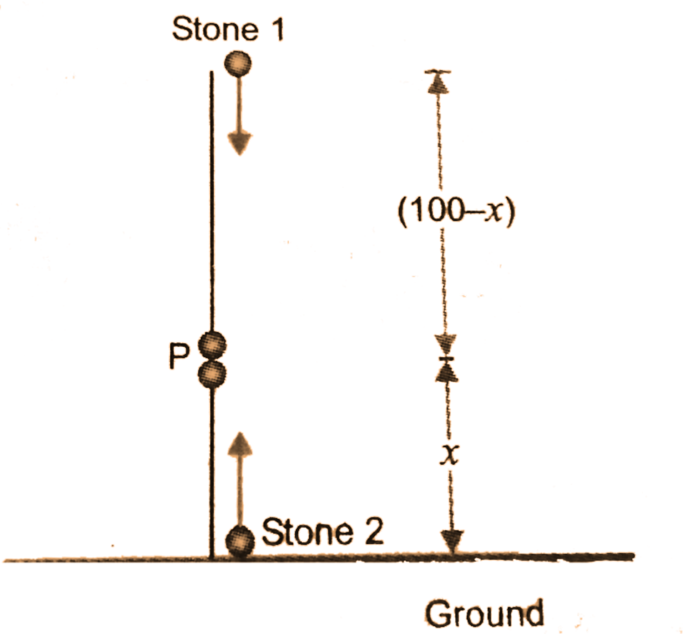
To find:

- Height, s=?

Solution:

- By third equation of motion
- v^2 - u^2 = 2gh
- 0 − 40^2 = −2 × 10 × h
- h = 1600 / 20
- ⇒h = 80m/s
- **Total distance travelled by stone = upward distance + downwards distance = 2 × h =160m**
- **Total Displacement =0, Since the initial and final point is same.**

12. A stone is allowed to fall from the top of a tower 100m high and at the same time another stone is projected vertically upwards from the ground with a velocity of 25m/s. Calculate when and where the two stones will meet.



- Let the stones meet at point A after time t.

For upper stone :

- u ′ = 0
- 100 - x = 0 + 1/2 g t^2

- 100 - x = 1/2 × 9.8 × t^2
- ⟹ 100 - x = 4.9 t^2 ....(1)

For lower stone :

- u = 25 m/s
- x = u t – ½ gt^2
- x = (25)t – 1/2 × 9.8× t^2
- ⟹x = 25 t – 4.9t^2 ....(2)

- Adding (1) and (2), we get
- 100 – x + x = 4.9 t^2 + 25 t - 4.9 t^2
- 100 = 25 t
- ⟹t=4 s
- From (1),
- 100 – x = 4.9 x 4 x 4
- 100 – x = 4.9 x 16
- 100 - x = 78.4
- x = 100 – 78.4
- ⟹x = 21.6 m

Hence the stone meet at a height of 21.6 m above the ground after 4 seconds.

13. A ball thrown Up vertically returns to the thrower after 6 seconds. Find i) velocity ii) Maximum height it reaches iii) Position after 4 seconds.

- Time to reach Maximum height ,
  t = 6/2 = 3 s. (since the ball thrown up vertically returns to the thrower in 6 seconds)
- v = 0 (at the maximum height)
- a = – 9.8 m/s^2

i) Using, v = u + gt, we get

- 0 = u – 9.8 × 3
- or, u = 29.4 ms^{-1}

Thus, the velocity with which it was thrown up = 29.4 m/s

ii) Using, 2ah = v^2 – u^2, we get

- h = v^2- u^2/2a

- $= 0 – 29.4 \times 29.4/- 2\times 9.8$
- $= 44.1$ m

Thus, Maximum height it reaches = 44.1 m

iii) t = 4s.

- In 3 s, the ball reaches the maximum height and in 1 s it falls from the top.
- Distance covered in 1 s from maximum height,
- $h = ut + 1/2\ gt^2$
- $= 0 + 1/2 \times 9.8 \times 1$
- $= 4.9$ m

From the bottom the maximum height reached will be = 44.1 – 4.9 = 39.2 m

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